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Extended release metformin for metabolic control assistance during prolonged clozapine administration: A 14 week, double-blind, parallel group, placebo-controlled study

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ABSTRACT

Background: Clozapine is the most effective agent in treatment-resistant schizophrenia. However, it is frequently associated with excessive body weight (BW) gain, type 2 diabetes mellitus and hyperlipidemia. The antidiabetic metformin (MET) has proved effective to assist in BW control during olanzapine administration. Therefore, we aimed to test whether MET may improve the metabolic profile in patients under prolonged clozapine administration.
Methods: In a double-blind, parallel group protocol, 61 patients (94.4% with schizophrenia) receiving clozapine (196.8 ± 132 mg daily, range: 25–500) for more than 3 consecutive months (86.5 ± 40.6 months, range: 4–168) were randomly allocated to extended release MET ($n = 31$; 500 to 1000 mg daily) or placebo ($n = 30$) group for 14 weeks. The BW, the body mass index, waist circumference, serum glucose, insulin, lipids, glycated hemoglobin (HBA1c), leptin and cortisol, and the HOMA-IR index were assessed at baseline, and weeks 7 and 14.
Results: MET was well tolerated and the mental state was not impaired during the study. The protocol was completed by all the placebo subjects and by 24 MET-treated patients. In a completer analysis at week 14, without including data of the 7 dropouts, the MET group lost − 1.87 ± 2.9 kg, whereas the placebo group had a stable BW: 0.16 ± 2.9 kg, $p = 0.01$ for the between group comparisons (effect size: 0.70). Leptin levels also tended to decrease after MET ($p = 0.08$). Insulin and the triglyceride-HDL-C ratio significantly decreased ($p < 0.05$, effect size 0.59 and 1.99 respectively) and the HDL-C significantly increased ($p = 0.001$, effect size 0.95) after MET.
Conclusions: MET improves metabolic control during prolonged clozapine administration.
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1. Introduction

The second generation antipsychotic drug (SGAD) clozapine is the most effective pharmacological agent in the treatment of refractory schizophrenia (Tandon et al., 2008). However, along with olanzapine, clozapine administration is associated with a high frequency of clinically-significant metabolic side effects (Newcomer, 2006). For instance, patients receiving clozapine up to 18 months in the phase 3 of the CATIE schizophrenia trial reported weight gain (l.3 lb per month), and increment in the blood levels of glucose (9.0 ± 34 mg/dL), triglycerides (55.6 ± 116.2 mg/dL) and glycosylated hemoglobin (HbA1c, 0.5 ± 1%) (Stroup et al., 2009). Besides, a 10-year follow-up of clozapine-treated patients

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reported estimate mortality from cardiovascular disease and new-onset diabetes mellitus of 9% and 43% respectively (Henderson et al., 2005a,b). Provided the well documented efficacy and effectiveness of clozapine in people with severe schizophrenia, proper assistance for metabolic control is needed in this overwhelmed clinical population (Newcomer, 2006).

Most pharmacological and non-pharmacological interventions for metabolic control during SGAD treatment (mainly olanzapine), have primarily focused on body weight gain (BWG) prevention or reduction and secondarily on improvement of carbohydrate and lipid metabolism (Baptista et al., 2008a,b; Faulkner et al., 2007; Citrome and Vreeland, 2008). Cognitive–behavioral training significantly prevented or reversed SGAD-induced BWG compared with standard care (Faulkner et al., 2007). The magnitude of body weight (BW) change for prevention was −3.38 kg (95% CI −4.8,−2.0) at medium term (16 weeks) and −4.87 kg (95% CI −7.1,−2.6) after 24 weeks. In the case of pharmacological studies for BW control during SGAD treatment conducted until the end of 2006 the reported figures were −1.16 kg (95% CI −1.9,−0.4) for prevention and −3.85 kg (95% CI −4.25,−3.44) for reversal, $p<0.05$ for all the above calculations.

Among the adjunctive drugs evaluated so far, the antidiabetic agent metformin (MET) appears particularly promising (Citrome and Vreeland, 2008). In the first place, considerable evidence has accumulated on MET use for BW and metabolic control in non-psychiatric populations. It is the first option for monotherapy in the treatment of type 2 diabetes mellitus. MET decreases glucose production, and improves glucose muscle uptake and the hepatic sensitivity to insulin (Spada and Philippe, 2008; Strack, 2008). Secondly, in psychiatric patients, an overall positive effect on BW, glucose/insulin homeostasis, lipid profile and leptin regulation have been reported since the year 2002 (Arman et al., 2008; Baptista et al., 2006, 2007a,b, 2008a,b; Chen et al., 2008; Klein et al., 2006; Morrison et al., 2002; Wu et al., 2008a,b). MET has been tested in clozapine-treated patients only in a 12-week, double-blind, placebo-controlled trial, with significant reduction in BW and the insulin resistance index (Wu et al., 2008b). However, the clozapine BW data was merged with that obtained from patients receiving olanzapine, risperidone or sulpiride and separated analysis for each SGAD was not provided.

We aimed to extend previous investigations by comparing MET and placebo in clozapine-treated subjects mainly affected of chronic and severe schizophrenia in a 14-week trial. The primary measure was the treatment impact on BW. Secondarily, we evaluated an insulin resistance index (HOMA-IR), blood lipids, HbA1c, serum cortisol and leptin levels. These two hormones are involved in BW and insulin sensitivity regulation, and cortisol is an indicator of the stress response (Baptista et al., 2007a). We did not set a threshold for BW or body mass index (BMI) for study entry, and expected a significant decrease in the anthropometric and biochemical variables in the MET group.

Distinctive features of this report were the use of extended release MET, which does not differ from the immediate release formulation in efficacy but displays less gastrointestinal side effects and is administered as a daily single dose (Schwartz et al., 2008). Besides we included the variables of non-high density lipoprotein cholesterol (non-HDL-C) and the triglyceride to high density lipoprotein cholesterol ratio (TG:HDL-C) recommended by Newcomer et al. (2008) as pointers of atherogenic and insulin resistance states respectively.

2. Methods

2.1. Subjects

The study was conducted during 2008 in Maracaibo city, Zulia State, Venezuela, at CATESFAM (Center for the Attention for Schizophrenia Outpatients and their Families). The protocol was approved by the Ethic's Committee of Los Andes University, Venezuela and a written informed consent was obtained from patients, caregivers and the head of the institution. The research team explicitly explained that MET might prevent or reverse clozapine-associated metabolic dysfunction, and the risk/benefit ratio was discussed in both situations. Subjects were invited for voluntary participation in the study without monetary compensation.

Since CATESFAM has educational programs, patients and relatives had already been sensitized about the metabolic side effects of clozapine, and all patients received recommendations for healthy life, which encouraged programmed physical activity and nutritional advice. However, the magnitude of compliance with this program and the changes in appetite were not formally quantified.

Inclusion criteria were to be under clozapine treatment for at least 3 consecutive months; to be older than 18 years, free of hormone replacement and have normal physical and laboratory tests (kidney, thyroid and liver). Elevated serum glucose and lipid levels were not considered as exclusion criteria. No subject had symptoms of type 2 diabetes; however, those with abnormal fasting glucose levels were carefully followed and were referred to an endocrinologist at the end of the study.

2.2. Study design and randomization

Sixty one clozapine-treated patients were randomly assigned by a computer program to either MET (Glucophage XR, Merck, Venezuela, 500 mg the first 2 weeks and then 1000 mg daily, $n=31$) or placebo ($n=30$) in identical pills. Both treatments were daily provided with the last meal in a double-blind protocol. Only the study coordinator (TB) knew the treatment assignment. Patients and caregivers attended several meeting where the study was formally explained. The clozapine and other psychotropic drug dosages were controlled by the treating psychiatrists and periodically informed to the study coordinator who adjusted the MET or placebo dose according to the individual tolerance.

Sample size was calculated based on an expected difference of BW change between the groups and standard deviation of 2 kg; a power of 80% and a type 1 error of 5%. The minimal size for each group was 16 subjects.

2.3. Assessments

Before starting the study, each patient was evaluated by a psychiatrist trained in clinical research that administered the Structured Clinical Interview for the DSMIV (First et al., 1997), the Brief Psychiatrist Rating Scale (BPRS) and verified the

inclusion criteria. The same psychiatrist assessed the mental state in every visit (baseline, weeks 7 and 14) and maintained contacts by phone with patients and caregivers every two weeks. Besides blood pressure (after 5 minute resting), the following variables were evaluated in strict fasting conditions.

2.3.1. Anthropometric variables

Body weight, height and waist circumference (WC, 1 cm above navel) were assessed in light clothes. The body mass index (BMI) was calculated as follows: BMI = weight (kg)/height squared.

2.3.2. Biochemical variables

Cubital vein blood samples were obtained. Glucose and lipids were quantified in every visit. Insulin, leptin and cortisol were only assessed at baseline and at week 14.

2.3.3. Chemical analysis

Glucose and lipids were measured with an enzymatic method from Humans (Germany). Insulin (normal values: 2–25 μIU/ml; detection limits: 1.76 μIU/ml), leptin (normal values: men: 3.84 ± 1.8 ng/ml; women: 7.36 ± 3.7; detection limits: 1.0 ng/ml) and cortisol (normal values: 5–22 μg/dl; detection limits: 0.25 μg/dl) were assessed by duplicate by ELISA with commercial kits from DRG, Germany. The inter- and intra-assay variation was below 10% for all assays. HbA1C levels were measured by a latex immunoagglutination inhibition method from DCA (Siemens, Canada; detection limits: 2.5%; reference values: 4.2–6.5%).

The insulin resistance index (HOMA-IR) was calculated as follows: glucose (mMol/dL) × insulin (mIU/mL)/22.4. The non-HDL-C was calculated by subtracting HDL-C from total cholesterol serum levels.

2.3.4. Statistical analysis

The SPSS 13.0 version (SPSS Inc. Chicago, Illinois, USA) was used. Data normality was estimated with the one-sample Kolmogorov–Smirnov test and Levene's test for the equality of variances.

For those variables assessed three times (basal, weeks 7 and 14) a repeated measures variance analysis was conducted. Within- and between group comparisons respectively were conducted with the two-tailed *t*-test for related and unrelated samples (for normally-distributed data), and with the Wilcoxon and Mann–Whitney *U* tests (for non-normally distributed data). Covariance analysis was conducted with basal BMI and HOMA-IR as covariates.

Bivariate correlation analysis was conducted with the Pearson and Spearman coefficients. Multivariate linear regression analysis was performed with the change in BW as dependent variable. Frequencies were analyzed with the chi-squared test. Results were considered significant at the $p < 0.05$ level.

The effect size for significant variables was calculated with the Cohen's *d* method (Cohen, 1988).

3. Results

All the placebo subjects ($n = 30$) completed the study. Seven out of 31 MET-treated patients abandoned the protocol before the first follow-up visit at week 7. Hence, a Last Observation Carried Forward (LOCF) analysis could not be achieved.

Table 1
Baseline demographic and clinical characteristics and pretreatment levels of outcome variables.

Variable	Group				Analysis	
	Clozapine plus metformin ($n = 24$)		Clozapine plus placebo ($n = 30$)			
	Mean	SD	Mean	SD	*t*	*p*
Brief Psychiatric Rating Scale	9.9	6.4	7.7	7.6	1.1	0.3
Age (yrs)	39.6	9.7	38.3	8.7	0.5	0.6
Duration of treatment (months)	98.4	40.4	78.9	39.6	1.7	0.1
Daily dose of clozapine (mg)	180.3	120.6	207.3	140.2	0.7	0.5
Body weight (kg)	82.2	20.9	77.1	15.8	1.0	0.3
Body mass index (kg/m^2)	28.7	5.3	27.4	5.7	0.8	0.4
Waist circumference (cm)	98.4	14.6	93.8	10.9	1.3	0.2
Glucose (mg/dl)	88.6	14.9	89.1	16.2	0.1	0.9
HbA1c (%)	5.5	0.4	5.4	0.3	1.1	0.3
Insulin (μIU/ml)	15.5	7.2	16.1	9.3	0.2	0.8
HOMA-IR	3.4	1.6	3.5	2.5	0.2	0.8
HDL cholesterol (mg/dl)	32.3	8.7	36.8	9.9	1.8	0.08
Non-HDL-C (mg/dl)	136.7	27.9	138.9	48.5	0.2	0.8
Triglycerides (mg/dl)	134.3	59.1	146.9	119.3	0.5	0.6
Triglycerides: HDL-C ratio (*)	4.6	0.6	4.6	0.88	0.9	0.3
Leptin (ng/ml)	25.1	21.0	22.2	21.2	0.5	0.6
Cortisol (pg/dl)	20.1	6.7	19.6	7.5	0.3	0.8
Systolic pressure (mm/hg)	121.4	12.1	121.9	16.1	0.1	0.9
Diastolic pressure (mm/hg)	75.5	7.8	75.9	11.8	0.1	0.9

One woman became pregnant, and the other dropouts had a severe mental disorder at baseline and manifested vague and unstable side effects that even the caregivers could not explain. None of these patients complained of gastrointestinal discomfort.

In the remainders, MET was well tolerated. One subject reported epigastric uneasiness which was managed by reducing the MET dose by half for 10 days. Then, the full dose was reinstated.

No significant differences were observed between the groups at baseline in any variable (Table 1).

Of the 54 subjects who completed the study, 51 had schizophrenia (45 paranoid schizophrenia [295.30] and 6 residual type [295.60], 2 had type I bipolar disorder [296.xx] and 1 had schizophreniform disorder [295.40]. Fifty subjects (93%) were mixed Hispanics and 4 (7%) were whites. The mental state as assessed by the BPRS remained stable along the study and no significant differences were observed *between* the treatment groups (Table 2).

3.1. Anthropometric variables

When participants in each treatment groups were classified as "obese" or "non-obese" at baseline (cut-off point, IMC > 27.5 kg/m^2), 50% in the MET group and 33.3% in the placebo group were classified as obese: χ^2 (1) = 1.5, $p = 0.2$.

Table 2
Change in relation to baseline in anthropometric and biochemical variables after metformin or placebo.

Variable	Treatment	Week 7			Week 14		
		Δ	Within-*t* (*p*)	Between-*t*(*p*)	Δ	Within-*t* (*p*)	Between-*t* (*p*)
BPRS	MET	−1.1 ± 2.9	1.4 (0.2)	0.7 (0.5)	−2.0 ± 5.9	1.7 (0.1)	0.8 (0.5)
	PBO	−1.9 ± 3.5	2.8 (0.08)		0.9 ± 5.9	0.9 (0.3)	
Body weight (kg)	MET	−0.87 ± 1.4	3.1 (0.01)	1.75 (0.08)	−1.87 ± 2.9	3.1 (0.01)	2.5 (0.01)
	PBO	0.08 ± 2.4	0.2 (0.8)		0.16 ± 2.9	0.3 (0.8)	
Body mass index (kg/m^2)	MET	−0.32 ± 0.4	3.1 (0.01)	1.8 (0.07)	−0.68 ± 1.0	3.3 (0.01)	2.6 (0.01)
	PBO	0.03 ± 0.8	0.2 (0.8)		0.05 ± 0.9	0.3 (0.7)	
Waist circumference (cm)	MET	−1.2 ± 2.9	2.1 (0.04)	0.1 (0.9)	−1.4 ± 3.5	1.9 (0.07)	1.2 (0.2)
	PBO	−1.1 ± 3.6	1.7 (0.09)		0.2 ± 3.6	0.3 (0.8)	
Glucose (mg/dl)	MET	−7.5 ± 13.8	2.6 (0.01)	0.9 (0.4)	−0.1 ± 18.4	0.1 (0.9)	0.5 (0.6)
	PBO	−4.3 ± 11.7	1.9 (0.05)		−2.2 ± 18.3	0.7 (0.5)	
Insulin (μIU/ml)	MET		–		−2.5 ± 3.7	3.2 (0.004)	2.1 (0.04)
	PBO				0.4 ± 5.8	0.1 (0.9)	
HOMA-IR	MET		–		−0.6 ± 1.1	2.4 (0.02)	0.9 (0.4)
	PBO				−0.1 ± 2.0	0.3 (0.7)	
Hb-A1c (%)	MET	−0.09 ± 0.2	2.0 (0.057)	2.1 (0.03)	0.13 ± 0.2	3.0 (0.006)	2.1 (0.04)
	PBO	0.03 ± 0.2	0.9 (0.4)		0.23 ± 0.14	9.1 (0.001)	
HDL cholesterol (mg/dl)	MET	5.5 ± 7.6	3.8 (0.001)	3.2 (0.002)	3.8 ± 7.5	2.7 (0.01)	3.5 (0.001)
	PBO	0.07 ± 4.6	0.4 (0.7)		−1.8 ± 3.7	2.8 (0.009)	
Non-HDL cholesterol (mg/dl)	MET	−7.4 ± 29.4	1.2 (0.2)	0.3 (0.7)	−12.9 ± 30.8	2.0 (0.052)	0.2 (0.8)
	PBO	−4.8 ± 35.6	0.7 (0.5)		−10.7 ± 36.9	1.6 (0.1)	
Triglycerides (*) (md/dl)	MET	11.7 ± 61.1	0.8 (0.4)	0.5 (0.6)	−4.1 ± 80.0	1.1 (0.3)	0.8 (0.4)
	PBO	26.8 ± 95.3	1.5 (0.1)		2.9 ± 82.4	0.4 (0.7)	
Triglycerides: HDL-C ratio (*)	MET	−0.4 ± 0.5	1.1 (0.2)	1.8 (0.07)	−0.8 ± 0.6	1.9 (0.04)	2.3 (0.02)
	PBO	1.2 ± 0.8	1.7 (0.09)		0.3 ± 0.5	0.9 (0.3)	
Systolic pressure (mm/hg)	MET	−6.3 ± 12.7	2.4 (0.02)	1.5 (0.1)	−4.7 ± 13.4	1.7 (0.1)	0.7 (0.5)
	PBO	−1.6 ± 9.9	0.9 (0.4)		−2.2 ± 10.7	1.1 (0.3)	
Diastolic pressure (mm/hg)	MET	−2.6 ± 8.3	1.5 (0.1)	1.5 (0.5)	−0.3 ± 6.5	0.8 (0.4)	1.1 (0.3)
	PBO	−1.5 ± 8.9	0.3 (0.8)		−1.5 ± 11.1	0.8 (0.5)	
Leptin (ng/ml)	MET		–		−5.4 ± 12.6	1.9 (0.06)	1.8 (0.08)
	PBO				0.26 ± 9.8	0.1 (0.9)	
Cortisol (pg/dl)	MET		–		−0.6 ± 4.8	0.5 (0.6)	0.4 (0.7)
	PBO				−1.1 ± 5.2	1.1 (0.3)	

(*) Between group comparisons: Mann Whitney *U* test; within group comparisons: Wilcoxon test.

A small but significant decrease was observed in BW and BMI after MET but not after placebo; for BW: (time: $f\ (2) = 3.5$, $p = 0.032$; time × treatment: $f\ (2) = 5.1$, $p = 0.008$); for the BMI: (time: $f\ (2) = 3.9$, $p = 0.02$; time × treatment: $f\ (2) = 5.7$, $p = 0.004$). The between group differences became significant at week 14 (effect size 0.70 and 0.76 for BW and BMI respectively, Table 2). Regarding the changes in BW, the basal BMI and the HOMA-IR were not significant covariates ($p > 0.05$).

While the WC tended to decrease after MET, there were no significant differences between the groups: (time: $f\ (2) = 3.3$, $p = 0.03$; time × treatment: $f\ (2) = 1.2$, $p = 0.3$), (see Table 2).

3.2. Biochemical variables

3.2.1. Glucose metabolism

The serum glucose levels tended to decrease over time but there were not significant differences between the groups: (time: $f\ (2) = 4.5$, $p = 0.01$; time × treatment: $f\ (2) = 0.9$, $p = 0.4$) (Table 2). Insulin levels and the HOMA-IR index significantly decreased in the MET group at week 14 ($p = 0.004$ and 0.02 respectively) but reached significance in the *between* group comparisons only for insulin levels (effect size 0.59) (Table 2).

The HbA1c levels displayed a marginally significant reduction after MET at week 7 and increased less with the antidiabetic at week 14 (effect size: 0.57) (Table 2). In the repeated measures analysis the figure reached marginal significance: time: $f\ (2) = 30$, $p = 0.001$; time × treatment: $f\ (2) = 2.7$, $p = 0.07$.

3.2.2. Blood lipids

The HDL-C levels significantly increased after MET and decreased after placebo: time: $f\ (2) = 6.6$, $p = 0.002$; time × treatment: $f\ (2) = 11.1$, $p = 0.001$, effect size: 0.95 (Table 2). The non-HDL-C levels changed over time but no differences were observed between the groups: time: $f\ (2) = 3.4$, $p = 0.03$; time × treatment: $f\ (2) = 0.05$, $p = 0.9$ (Table 2). When considering the changes in HDL-C, the basal BMI and the HOMA-IR were not significant covariates ($p > 0.05$).

No significant changes were observed in triglyceride levels: time and time × treatment effect: $p > 0.05$ (Table 2). However, the TG:HDL-C ratio showed a small but significant decrease at week 14 after MET, ($p = 0.02$, effect size: 1.99, Table 2).

3.2.3. Blood pressure

Systolic pressure decreased after time ($p = 0.04$) particularly in the MET group (Table 2) but no *between* group differences were observed (time × treatment effect, $p = 0.2$ (Table 2).

No significant changes were observed in the diastolic pressure: time effect, $p=0.5$; time × treatment, $p=0.3$ (Table 2).

3.2.4. Leptin and cortisol

The leptin levels tended to decrease in the MET group ($p=0.06$) but the between group comparison only reached marginally significance ($p=0.08$) (Table 2). The same trend for leptin reduction in the MET group was observed when controlling by gender: women: $p=0.1$; men: $p=0.2$ (data not shown).

The cortisol levels remained stable along the study with no significant differences in the within- or between group comparisons (Table 2).

3.2.5. Frequency of the metabolic syndrome

The frequency of the metabolic syndrome defined according to the Cholesterol Panel Criteria (Third Report of the National Cholesterol Education Program, 2002) was 25% and 16.7% respectively before and after MET administration: κ^2 (1) = 0.51, $p=0.4$, whereas it remained identical (16.7%) before and after placebo κ^2 (1) = 0.00, $p=1.0$.

An additional analysis was to compare the change in the variables under study according to the history of type 2 diabetes in first-degree relatives. Within each treatment group, no significant difference was observed in any anthropometric or biochemical variable in subjects with or without family history (data not shown).

3.2.6. Correlation and regression analysis

We focused on the change in BW, the HOMA-IR index and HDL-C levels since these were the variables that attained statistical significance (Table 2).

3.2.6.1. Change in BW. In the bivariate correlation analysis the change of BW at week 14 did not correlate significantly with the basal BPRS, anthropometric or biochemical variables or gender in any treatment group ($p>0.05$, data not shown). Within each treatment group, the BW change was similar in women and men ($p>0.05$, data not shown).

The change in BW significantly correlated with the change in WC and HOMA-IR in the MET group: $r=0.59$, $p=0.002$ and $r=0.43$, $p=0.04$ respectively. In the placebo group, the change in BW significantly correlated with the change in leptin levels ($r=0.4$, $p=0.03$); triglyceride levels ($r=0.58$, $p=0.001$); TG:HDL-C ratio ($r=0.53$, $p=0.003$) and with the change in WC ($r=0.81$, $p=0.0001$).

In the multivariate regression analysis taking the BW change as dependent variable and the change in the other clinical, anthropometric and biochemical as independent ones at day 14, besides the expected positive association with the change in the BMI ($p=0.001$ in both groups), only the change in the diastolic pressure reached significance in the placebo group: standardized β-coefficient = −0.047, $t=-2.3$, $p=0.044$.

3.2.6.2. Change in the HOMA-IR index. In the bivariate correlation analysis, the change in the HOMA-IR index at week 14 correlated negatively with the basal levels of insulin and glucose in both groups: insulin: MET group: $r=-0.4$, $p=0.06$; placebo group: $r=-0.5$, $p=0.01$; glucose: MET group: $r=-0.6$, $p=0.006$; placebo group: $r=-0.6$, $p=0.001$. The change in the HOMA-IR index at week 14 showed the expected positive correlations with the changes in insulin and glucose in both treatment groups ($p<0.001$ in all comparisons) and with the changes in WC and BMI only in the MET group ($r=0.4$, $p=0.04$ for both variables). In the placebo group, the change in the HOMA-IR correlated positively with the change in cortisol levels: $r=0.39$, $p=0.04$.

In the multivariate regression analysis taking the change in the HOMA-IR index as dependent variable and the change in the other clinical, anthropometric and biochemical as independent ones at day 14, the expected significant associations with the changes in insulin were observed: standardized β-coefficient = 0.73, $t=2.5$, $p=0.049$ and 1.02, $t=12.05$, $p=0.001$ for the MET and placebo group respectively. In this last group, a positive association was also observed with the change in triglycerides: β-coefficient = 0.21, $t=2.5$, $p=0.027$.

3.2.6.3. Change in HDL-C. In the bivariate correlation analysis the change in HDL-C at week 14 only showed a marginally significant correlation with basal insulin ($r=41$, $p=0.051$) and with the basal HOMA-IR ($r=0.42$, $p=0.053$) in the MET group. No significant correlation was observed in the placebo group. The change in HDL-C did not correlate significantly with the change in any other variable in the MET group, but showed a negative association with the change in cortisol levels in the placebo group: $r=-0.38$, $p=0.04$.

In the multivariate regression analysis taking the change in HDL-C as dependent variable and the change in the other clinical, anthropometric and biochemical as independent ones at day 14, no variable displayed significant association in the MET group ($p>0.05$, data not shown). In the placebo group, a positive association was observed with the change in non-HDL-C: standardized β-coefficient = 0.53, $t=2.5$, $p=0.029$.

4. Discussion

With the exception of a case report (Ozeroglu et al., 2007) and the study of Wu et al. (2008b), we did not find a placebo-controlled published trial of metformin in clozapine-treated subjects. Other clozapine studies have evaluated phenylpropanolamine (Borovicka et al., 2002); the ling gui zhu gan tang mixture (Ding et al., 2005); modafinil (Henderson et al., 2005a,b); sibutramine (Henderson et al., 2007); topiramate (Ko et al., 2005) and fluvoxamine (Lu et al., 2004). Hence, we did not count on additional data to be directly compared with our results. In fact, most published studies on MET in adult psychiatric patients involved olanzapine. Besides, in contrast with the present trial, some studies were designed to assess the effect of MET to *prevent* olanzapine-induced BWG (Baptista et al., 2006; Wu et al., 2008a) or combined MET with sibutramine (Baptista et al., 2008a). With these limitations in mind, the impact of MET on the diverse variables of the metabolic syndrome during SGADs will be discussed separately. A brief overview of MET effects in non-psychiatric populations will be first offered.

4.1. MET in non-psychiatric populations

While the positive impact of MET on glucose and insulin homeostasis in non psychiatric populations is well established (Spada and Philippe, 2008; Strack, 2008), its effects on BW and lipid profile are more controversial. Considering BW,

a recent review evaluated 5 of 12 trials in adults taking BW loss as a primary endpoint. Significant weight reduction was found in 4 of these studies (Desilets et al., 2008). Thus, MET appears promising in the assistance of BW control, and in some subjects it displays a potent anorectic effect (Helvaci et al., 2008).

On the other hand, a systematic review of randomized-controlled clinical trials suggests that MET has no intrinsic effect on blood pressure, HDL-C and triglycerides in patients with type 2 diabetes (Wulffelé et al., 2004). However, MET appears to reduce total and LDL-C independently of its effects on glucose control (Wulffelé et al., 2004). A recent study showed that early intervention with intensive lifestyle interventions but less so with MET in subjects at high risk for diabetes improves the cardiovascular risk and glucose tolerance profile simultaneously (The Diabetes Prevention Program Research Group, 2009).

Lastly, an *in vitro* study reported that metformin increased intracellular and extracellular beta-amyloid peptides, which are involved in the pathogenesis of Alzheimer disease (Chen et al., 2009). However, the clinical relevance of this finding waits for additional studies.

4.2. Body weight

Previous studies in patients after prolonged olanzapine (Baptista et al., 2007b) or several SGAD including clozapine and olanzapine (Wu et al., 2008b) administration reported the following BW loss (in Kg) for the MET and placebo group respectively (the trial duration is also mentioned): Baptista et al. (2007): -1.4 ± 3.2 vs. -0.18 ± 2.8 after 12 weeks; Wu et al. (2008b); -3.2; 95% CI, 2.5–3.9 vs. $+3.1$; 95% CI, 2.4–3.8 after 12 weeks.

Hence, the magnitude of BW change reported here (-1.87 ± 2.9 vs. 0.16 ± 2.9, with a medium effect size of 0.70 according to Cohen's ranking) was close to that observed in our report of MET administration after olanzapine treatment (Baptista et al., 2007b). Importantly, in this study, clozapine had been administered during 86.5 months consecutively in the whole sample (range: 4–164 months). Since after this prolonged treatment BW is expected to be stable in a severely ill population, the relatively small BW decrement observed in our study may be clinically significant.

4.3. Glucose homeostasis

In the previous study with olanzapine-treated subjects, the HOMA-IR remained stable in the MET group (-0.09 ± 3.1, $p = 0.8$) and significantly increased after placebo (0.91 ± 1.7, $p = 0.006$) (Baptista et al., 2007b). In the study of Wu et al. (2008b) the change was more pronounced: MET: -3.5, CI 95%: -4.4 to -2.7; placebo: 0.4, CI 95%, -0.1 to -0.7. In the present study, the HOMA-IR showed a small but significant decrease after MET: -0.9 ± 1.1 ($p = 0.02$) vs. -0.1 ± 2.0 ($p = 0.7$) after placebo. The positive impact of MET was based on its effect on insulin (with a medium effect size of 0.59) instead of decreasing glucose levels, without a significant effect on the HOMA-IR index (Table 2). This may be related to the fact that most subjects did not have abnormal fasting glucose but relatively high insulin levels at baseline (Table 1). The relative stability of the HOMA-IR index in the placebo group contrasts with the significant increase after placebo in the previous studies, perhaps due to the already discussed difference in treatment duration.

The HbA1c levels were not assessed by Wu et al (2008b). During olanzapine treatment, MET did not prevent an additional increase in HbA1c levels, although the figure did not reach significance in the between group comparison (0.47 ± 0.1 vs. 0.42 ± 1.5, $p = 0.8$) (Baptista et al., 2007). In the present study, while the HbA1c levels increased in both groups, the increment was significantly lower after MET (0.13 ± 2.0 vs. 0.23 ± 0.14, $p = 0.04$, effect size 0.59).

4.4. Blood lipids

MET improved the lipid profile by significantly increasing HDL-C levels and decreasing the TG:HDL-C ratio (Table 2). For both variables, the effect size reached the highest values in this study (0.95 and 1.99 respectively). During olanzapine administration a positive effect of MET was not observed, since HDL-C levels tended to decrease: (-5.8 ± 12 vs. -2.0 ± 13.9 mg/dl, $p = 0.2$) (Baptista et al., 2007b). Lipids levels were not reported by Wu et al. (2008b).

4.5. Blood pressure

Blood pressure changes were not reported by Baptista et al. (2007b) and Wu et al. (2008b). In the present study, while a significant reduction of systolic pressure in the MET group was observed at week 7 ($p = 0.02$), no between group differences was observed at any time point (Table 2).

4.6. Leptin and cortisol levels

A marginally significant decrease in leptin levels was observed after MET ($p = 0.06$), while values remained stable after placebo ($p = 0.08$ in the between group comparison) (Table 2). Since leptin is a pointer of body adiposity, this result is consistent with the observed decrease in BW. No significant changes in cortisol levels were observed in any treatment group (Table 2). Similar results on leptin and cortisol were reported in our previous study in olanzapine-treated subjects (Baptista et al., 2007b). None of these hormones were assessed by Wu et al. (2008b).

Leptin and cortisol were measured here in an exploratory way. Preliminary clinical and experimental evidence suggest that MET decreases leptin synthesis and response to BW changes. Baptista et al. (2007a) showed that in spite of a significant olanzapine-induced BW gain, MET prevented the expected increase in leptin levels in schizophrenia patients. In the present study, the predictable positive correlation between BW change and leptin levels was observed after placebo (r [27] $= 0.41$, $p = 0.03$) but only reached marginal significance after MET (r [22] $= 0.37$, $p = 0.09$). This may be an indirect pointer of the modulating effect of the antidiabetic agent on leptin homeostasis, which must be assessed in future studies.

Provided the positive effects of MET on insulin sensitivity, we expected a decrease in cortisol levels, which is an insulin counter regulatory- and stress-response hormone. This secondary hypothesis was not confirmed here and must be explored in future studies with a different methodology.

5. Limitations and conclusions

This study comprised subjects within a wide range of BMI and metabolic variables. Hence, it must be considered as a mixed protocol of prevention and reversal of clozapine-induced metabolic dysfunction. Besides, the BW change after initiating clozapine could not be assessed due to the long time elapsed since the treatment start, lack of proper early BW assessment, and severity of the mental disorder. And finally, we could not assess the impact on the results of the 7 drop-outs in the MET group.

While this trial showed a positive effect of MET future studies must be conducted in subjects with more homogeneous BMI, metabolic parameters and age. Besides, it must be clarified whether ethnic differences influence the MET effects on BW and metabolism. For example, studies conducted in Chinese populations (Wu et al., 2008a,b) reported higher rates of BW loss compared to our studies in more mixed populations (Baptista et al., 2006, 2007b, 2008a). And lastly, higher MET doses and more prolonged administration, this last point as suggested by Faulkner et al. (2007) must be considered.

An issue of the greatest importance is when to start MET administration and in which specific subjects. An intuitive preliminary approach is that besides non-pharmacological measures which may be very effective (Citrome and Vreeland, 2008) MET could be provided early in treatment in patients with personal and/or family history of obesity, type 2 diabetes, hyperlipidemia and abnormal pointers of insulin sensitivity (HOMA-IR index, TG-HDL-C ratio, etc). More refined criteria that are under investigation are, for instance, the presence of lower or higher than expected serum leptin levels in relation to the individual BMI or body fat percentage, or specific leptin receptor polimorphisms (Peña et al., 2008).

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Contributors

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Conflict of interest

None.

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